

synthetic

mixed STEM responses

Q1 $u = x^2, du = 2x dx$

$$I = \frac{1}{2} \int_0^1 e^u du = \frac{1}{2}(e - 1)$$

Q2 ~~$\mu = 0.20$~~ ~~$N = mg \cos 30^\circ$~~ OCR: μ , not u

$$a = g(\sin 30^\circ - \mu \cos 30^\circ) = 3.20$$

$$v = \sqrt{2as} = 4.38 \text{ m/s}$$

Q3 If $A^T Ax = 0$, then $Ax = 0$ justify with $\|Ax\|^2$
because A^T cancels A .

Q4

$$m = \max(xs)$$

$$z = [\exp(x-m) \text{ for } x \text{ in } xs]$$

$$\text{return } [v / \text{sum}(z) \text{ for } v \text{ in } z]$$

empty input?